



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI LINGESH AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

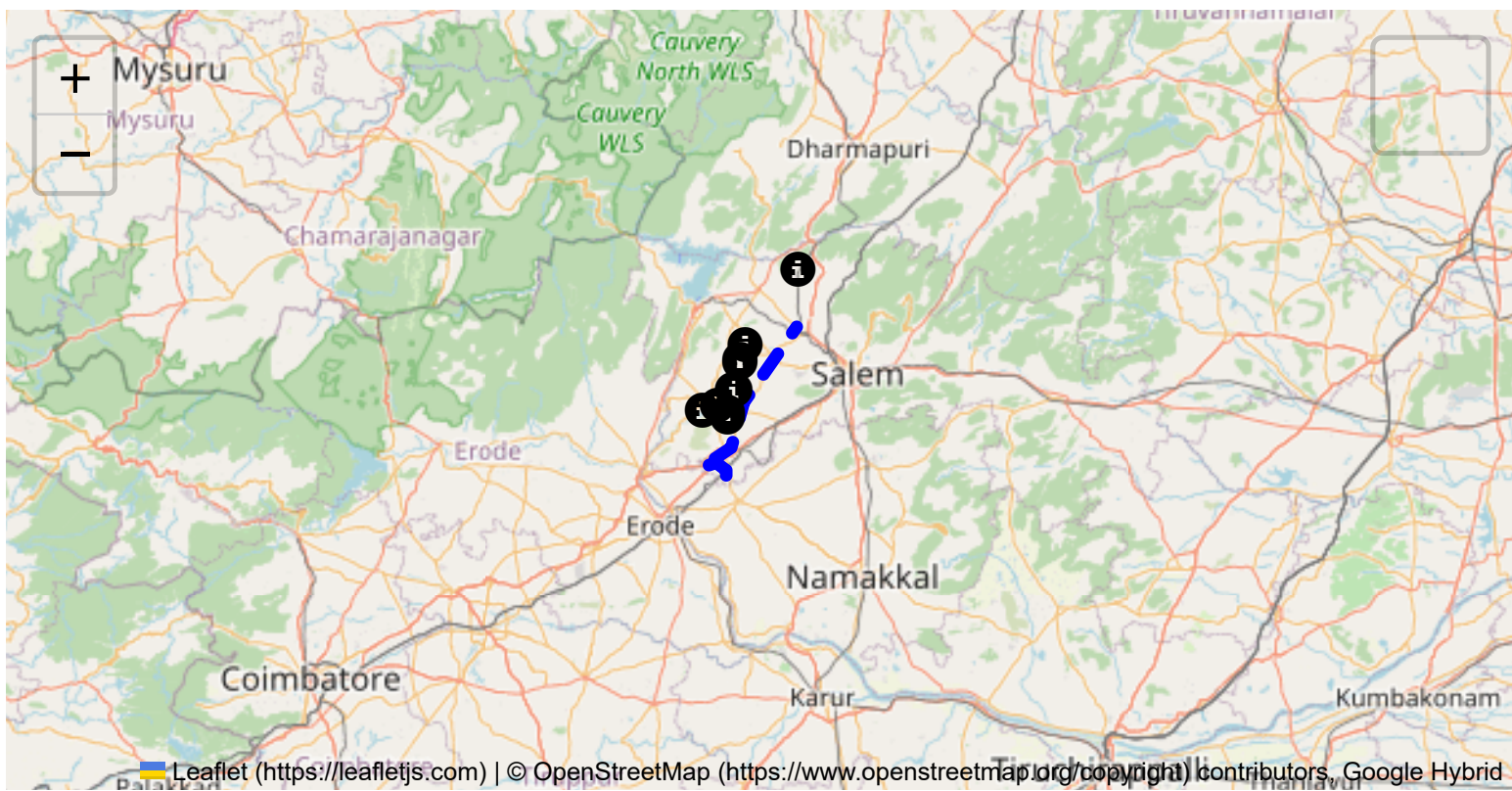
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

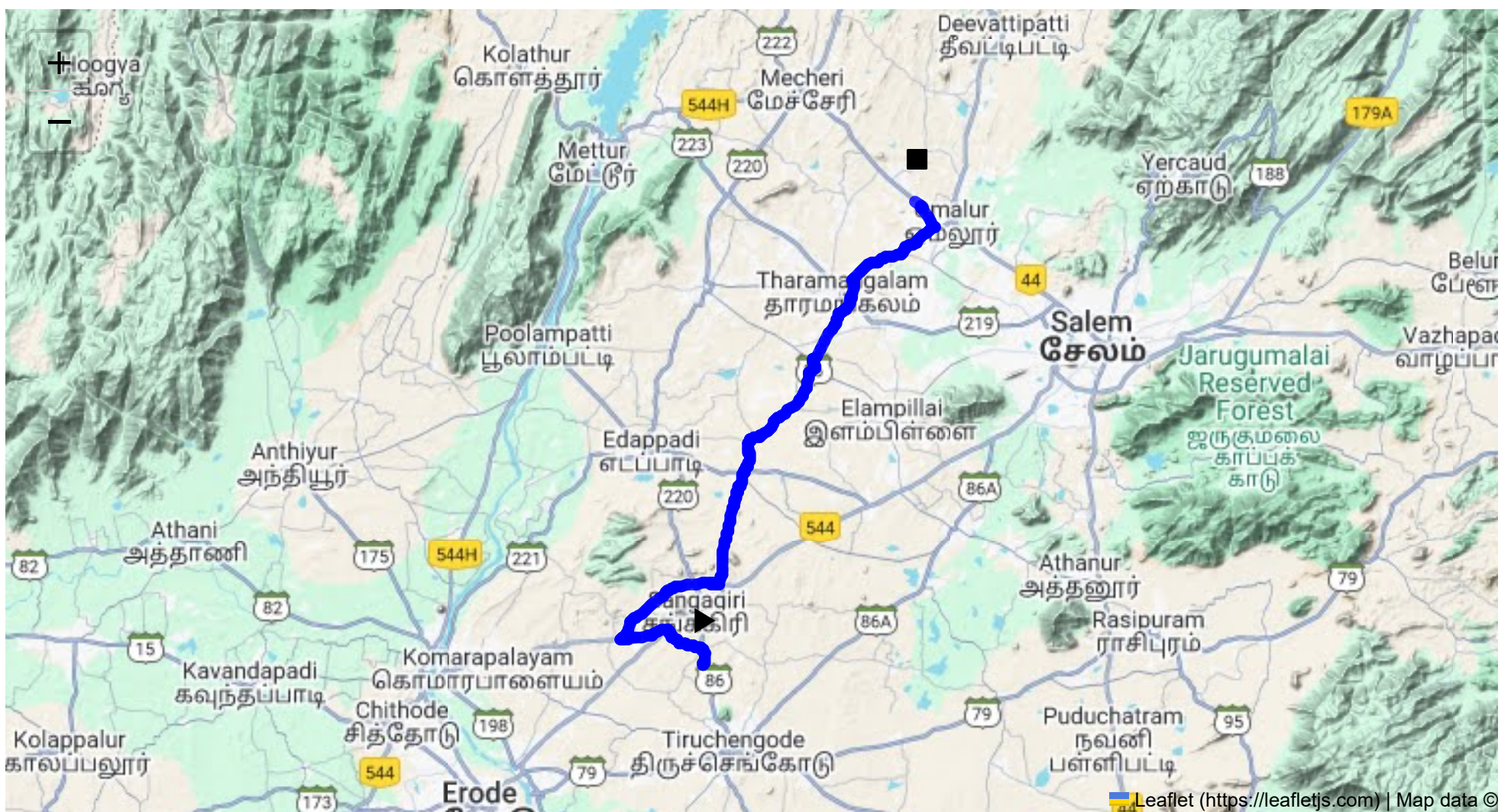
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 54.88 km
Estimated Duration: 1.3 hours
Adjusted Duration (Heavy Vehicle): 1.7 hours
Start: (11.4381, 77.8734)
End: (11.755047, 78.022331)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Omalur via specified waypoints in Tamil Nadu, India, spans approximately 54.88 kilometers. It includes a combination of local roads and state highways, with State Highway 86 (SH 86) being a significant portion of the travel path.

Typical Weather Conditions and Potential Hazards

- **Climate:** The region typically experiences a tropical climate with a hot and dry summer, monsoon season from June to September, and a mild winter.
- **Weather Hazards:** Monsoon can result in heavy rainfall, causing road flooding and slippery conditions. Fog during winter mornings may affect visibility.

Traffic Patterns

- **Peak Hours:** Traffic tends to peak during the early morning (8-10 AM) and late afternoon (5-7 PM) due to local commuter patterns.
- **Congestion-Prone Areas:** Key congestion points are usually found near major junctions such as Sankari and Erumaipatti, where local and through traffic intersects.

Road Quality and Infrastructure

- **Assessment:** Road surfaces are varied; local roads may have uneven patches and require cautious driving, while SH 86 is generally well-maintained but can experience occasional potholes.
- **Infrastructure:** Adequate road signage, although illumination may be poor on less urban stretches.

Alternative Routes for Emergencies

- **Suggestions:** In case of obstruction, Route SH 20 can act as an alternative path connecting near Erumaipatti to bypass SH 86.

Local Regulations for Hazardous Material Transport

- **Regulations:** Compliance with state transport regulations is vital. Permits are required for hazardous materials, and movement is typically controlled during late evening to early morning hours to minimize risk.

Historical Incidents

- **Overview:** There have been a few incidents involving vehicles carrying hazardous materials, primarily due to driver error or adverse weather. No major environmental disasters reported in recent history.

Environmental Considerations

- **Sensitive Areas:** Close proximity to agricultural fields and neighborhoods requires careful navigation of materials to prevent spills and contamination.

Communication Coverage

- **Dead Zones:** Sporadic mobile network coverage, especially in rural stretches between Erumaipatti and Koranampatti. Emergency communication devices are recommended.

Estimated Emergency Response Times

- Response Times:** Typically, emergency services can be expected to reach most parts of the route within 30-45 minutes, depending on traffic and weather conditions.

Overall Risk Assessment Summary

The route poses moderate risk factors primarily due to varied road conditions, traffic congestion at certain points, and weather-related challenges during the monsoon season. Proper adherence to transport regulations, strategic planning of travel times, and real-time communication with dispatch centers will mitigate most risks. Regular route inspections and contact with local authorities are also recommended to keep abreast of new developments or regulations impacting transport safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.45357, 77.85789	30 KM/Hr
4	Turn	Medium	11.45352, 77.85698	30 KM/Hr
5	Turn	Medium	11.46048, 77.85036	30 KM/Hr
6	Turn	Medium	11.46303, 77.84945	30 KM/Hr
7	Turn	Medium	11.46314, 77.84928	30 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Turn	High	11.45634, 77.81673	15 KM/Hr
10	Turn	Medium	11.49421, 77.88021	30 KM/Hr
11	Turn	High	11.49383, 77.88526	15 KM/Hr
12	Turn	High	11.49395, 77.88537	15 KM/Hr
13	Turn	High	11.49409, 77.88536	15 KM/Hr
14	Turn	Medium	11.49435, 77.88499	30 KM/Hr
15	Turn	Medium	11.49465, 77.88488	30 KM/Hr
16	Turn	Medium	11.56072, 77.89919	30 KM/Hr
17	Turn	Medium	11.56096, 77.89931	30 KM/Hr
18	Turn	High	11.57035, 77.90223	15 KM/Hr
19	Turn	Medium	11.57041, 77.90221	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
20	Turn	Medium	11.58279, 77.90658	30 KM/Hr
21	Turn	Medium	11.64197, 77.95197	30 KM/Hr
22	Turn	Medium	11.67858, 77.96834	30 KM/Hr
23	Turn	High	11.68915, 77.97734	15 KM/Hr
24	Turn	Medium	11.70218, 77.97866	30 KM/Hr
25	Turn	High	11.70227, 77.97864	15 KM/Hr
26	Turn	Medium	11.71921, 78.01133	30 KM/Hr
27	Blind Spot	Blind Spot	11.73785, 78.03551	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46048, 77.85036



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45634, 77.81673



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49421, 77.88021



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88526



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49395, 77.88537



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49409, 77.88536



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49435, 77.88499



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56072, 77.89919



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56096, 77.89931



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57035, 77.90223



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57041, 77.90221



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58279, 77.90658



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64197, 77.95197



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.68915, 77.97734



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70218, 77.97866



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71921, 78.01133



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.73785, 78.03551
